

ApexPlanet — Task 4

Exploitation & System Security

Controlled Metasploitable2 Lab Assessment

Assessment target: Metasploitable2 — 192.168.56.102

Assessment VM: Kali Linux — 192.168.56.101 (lab interface)

Scope: Private VirtualBox lab environment

Executive Summary

This report documents a controlled assessment of the intentionally vulnerable Metasploitable2 virtual machine. The evidence covers reconnaissance, service/version discovery, vulnerability assessment, SSH security review, system-hardening checks, phishing awareness simulation, and a benign malware-analysis exercise.

The clearest documented vulnerability is VSFTPD 2.3.4. Metasploit was used to identify the corresponding VSFTPD 2.3.4 backdoor module and review its configuration. The captured evidence does not claim successful remote-shell or post-exploitation access.

1. Scope and Methodology

The assessment was performed only against the deliberately vulnerable lab target on the private VirtualBox network. The workflow consisted of host discovery, service/version enumeration, vulnerability identification, configuration review, hardening-state checks, and controlled awareness/analysis exercises.

Lab addresses observed in the evidence

System	Address	Role
Kali Linux	192.168.56.101	Assessment VM
Metasploitable2	192.168.56.102	Intentionally vulnerable target

2. Reconnaissance and Service Discovery

Host discovery located the Metasploitable2 VM on the lab subnet. A version/OS scan then identified a broad exposed service surface, including FTP, SSH, Telnet, SMTP, DNS, HTTP, SMB, RPC, databases, VNC, IRC, AJP and Tomcat.

Reconnaissance / host discovery

```
kali@kali: ~
Session Actions Edit View Help
Docs: nmapshd(8)
man:ssh config(5)

kali@kali:~$ nmap -v -sS 192.168.56.102
Starting Nmap 7.90 ( https://nmap.org ) at 2020-09-11 05:09 -0400
Nmap scan report for 192.168.56.102
Host is up (0.11s latency).
Not shown: 979 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian subuntul (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache/2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC B100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rshexec
513/tcp   open  login        Netkit rshd
514/tcp   open  shell        Netkit rshd
1699/tcp  open  java-rmi     GNU Classpath gmrregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC B100000)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a Ubuntu5
5432/tcp  open  postgresql   PostgreSQL UB 8.3.0 - 8.3.7
5900/tcp  open  vnc           VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
8080/tcp  open  irc          UnrealIRCd
8089/tcp  open  ajp13        Apache/2.2.8 (Protocol V1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSF engine 1.1
MAC Address: 08:00:27:82:32:BE (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.x
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network distance: 1 hop
Service Info: hosts: metasploitable, localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service Detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 15.59 seconds

kali@kali:~$
```

Service and version discovery overview

```
kali@kali: ~
Session Actions Edit View Help

CVE-2019-12035 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12035 *EXPLOIT*
CVE-2019-12036 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12036 *EXPLOIT*
CVE-2019-12037 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12037 *EXPLOIT*
CVE-2019-12038 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12038 *EXPLOIT*
CVE-2019-12039 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12039 *EXPLOIT*
CVE-2019-12040 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12040 *EXPLOIT*
CVE-2019-12041 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12041 *EXPLOIT*
CVE-2019-12042 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12042 *EXPLOIT*
CVE-2019-12043 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12043 *EXPLOIT*
CVE-2019-12044 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12044 *EXPLOIT*
CVE-2019-12045 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12045 *EXPLOIT*
CVE-2019-12046 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12046 *EXPLOIT*
CVE-2019-12047 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12047 *EXPLOIT*
CVE-2019-12048 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12048 *EXPLOIT*
CVE-2019-12049 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12049 *EXPLOIT*
CVE-2019-12050 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12050 *EXPLOIT*
CVE-2019-12051 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12051 *EXPLOIT*
CVE-2019-12052 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12052 *EXPLOIT*
CVE-2019-12053 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12053 *EXPLOIT*
CVE-2019-12054 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12054 *EXPLOIT*
CVE-2019-12055 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12055 *EXPLOIT*
CVE-2019-12056 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12056 *EXPLOIT*
CVE-2019-12057 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12057 *EXPLOIT*
CVE-2019-12058 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12058 *EXPLOIT*
CVE-2019-12059 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12059 *EXPLOIT*
CVE-2019-12060 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12060 *EXPLOIT*
CVE-2019-12061 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12061 *EXPLOIT*
CVE-2019-12062 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12062 *EXPLOIT*
CVE-2019-12063 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12063 *EXPLOIT*
CVE-2019-12064 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12064 *EXPLOIT*
CVE-2019-12065 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12065 *EXPLOIT*
CVE-2019-12066 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12066 *EXPLOIT*
CVE-2019-12067 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12067 *EXPLOIT*
CVE-2019-12068 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12068 *EXPLOIT*
CVE-2019-12069 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12069 *EXPLOIT*
CVE-2019-12070 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12070 *EXPLOIT*
CVE-2019-12071 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12071 *EXPLOIT*
CVE-2019-12072 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12072 *EXPLOIT*
CVE-2019-12073 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12073 *EXPLOIT*
CVE-2019-12074 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12074 *EXPLOIT*
CVE-2019-12075 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12075 *EXPLOIT*
CVE-2019-12076 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12076 *EXPLOIT*
CVE-2019-12077 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12077 *EXPLOIT*
CVE-2019-12078 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12078 *EXPLOIT*
CVE-2019-12079 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12079 *EXPLOIT*
CVE-2019-12080 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12080 *EXPLOIT*
CVE-2019-12081 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12081 *EXPLOIT*
CVE-2019-12082 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12082 *EXPLOIT*
CVE-2019-12083 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12083 *EXPLOIT*
CVE-2019-12084 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12084 *EXPLOIT*
CVE-2019-12085 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12085 *EXPLOIT*
CVE-2019-12086 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12086 *EXPLOIT*
CVE-2019-12087 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12087 *EXPLOIT*
CVE-2019-12088 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12088 *EXPLOIT*
CVE-2019-12089 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12089 *EXPLOIT*
CVE-2019-12090 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12090 *EXPLOIT*
CVE-2019-12091 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12091 *EXPLOIT*
CVE-2019-12092 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12092 *EXPLOIT*
CVE-2019-12093 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12093 *EXPLOIT*
CVE-2019-12094 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12094 *EXPLOIT*
CVE-2019-12095 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12095 *EXPLOIT*
CVE-2019-12096 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12096 *EXPLOIT*
CVE-2019-12097 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12097 *EXPLOIT*
CVE-2019-12098 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12098 *EXPLOIT*
CVE-2019-12099 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12099 *EXPLOIT*
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CVE-2019-12102 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12102 *EXPLOIT*
CVE-2019-12103 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12103 *EXPLOIT*
CVE-2019-12104 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12104 *EXPLOIT*
CVE-2019-12105 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12105 *EXPLOIT*
CVE-2019-12106 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12106 *EXPLOIT*
CVE-2019-12107 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12107 *EXPLOIT*
CVE-2019-12108 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12108 *EXPLOIT*
CVE-2019-12109 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12109 *EXPLOIT*
CVE-2019-12110 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12110 *EXPLOIT*
CVE-2019-12111 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12111 *EXPLOIT*
CVE-2019-12112 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12112 *EXPLOIT*
CVE-2019-12113 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12113 *EXPLOIT*
CVE-2019-12114 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12114 *EXPLOIT*
CVE-2019-12115 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12115 *EXPLOIT*
CVE-2019-12116 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12116 *EXPLOIT*
CVE-2019-12117 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12117 *EXPLOIT*
CVE-2019-12118 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12118 *EXPLOIT*
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CVE-2019-12120 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12120 *EXPLOIT*
CVE-2019-12121 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12121 *EXPLOIT*
CVE-2019-12122 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12122 *EXPLOIT*
CVE-2019-12123 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12123 *EXPLOIT*
CVE-2019-12124 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12124 *EXPLOIT*
CVE-2019-12125 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12125 *EXPLOIT*
CVE-2019-12126 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12126 *EXPLOIT*
CVE-2019-12127 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12127 *EXPLOIT*
CVE-2019-12128 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12128 *EXPLOIT*
CVE-2019-12129 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12129 *EXPLOIT*
CVE-2019-12130 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12130 *EXPLOIT*
CVE-2019-12131 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12131 *EXPLOIT*
CVE-2019-12132 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12132 *EXPLOIT*
CVE-2019-12133 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12133 *EXPLOIT*
CVE-2019-12134 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12134 *EXPLOIT*
CVE-2019-12135 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12135 *EXPLOIT*
CVE-2019-12136 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12136 *EXPLOIT*
CVE-2019-12137 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12137 *EXPLOIT*
CVE-2019-12138 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12138 *EXPLOIT*
CVE-2019-12139 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12139 *EXPLOIT*
CVE-2019-12140 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12140 *EXPLOIT*
CVE-2019-12141 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12141 *EXPLOIT*
CVE-2019-12142 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12142 *EXPLOIT*
CVE-2019-12143 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12143 *EXPLOIT*
CVE-2019-12144 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12144 *EXPLOIT*
CVE-2019-12145 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12145 *EXPLOIT*
CVE-2019-12146 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12146 *EXPLOIT*
CVE-2019-12147 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12147 *EXPLOIT*
CVE-2019-12148 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12148 *EXPLOIT*
CVE-2019-12149 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12149 *EXPLOIT*
CVE-2019-12150 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12150 *EXPLOIT*
CVE-2019-12151 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12151 *EXPLOIT*
CVE-2019-12152 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12152 *EXPLOIT*
CVE-2019-12153 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12153 *EXPLOIT*
CVE-2019-12154 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12154 *EXPLOIT*
CVE-2019-12155 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12155 *EXPLOIT*
CVE-2019-12156 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12156 *EXPLOIT*
CVE-2019-12157 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12157 *EXPLOIT*
CVE-2019-12158 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12158 *EXPLOIT*
CVE-2019-12159 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12159 *EXPLOIT*
CVE-2019-12160 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12160 *EXPLOIT*
CVE-2019-12161 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12161 *EXPLOIT*
CVE-2019-12162 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12162 *EXPLOIT*
CVE-2019-12163 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12163 *EXPLOIT*
CVE-2019-12164 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-12164 *EXPLOIT*
CVE-2019-12165 2019-12-03 10.0 https://vulners.com/gitee/CVE-2019-1216
```

```
Session Actions Edit View Help
Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapidly Open Source Project

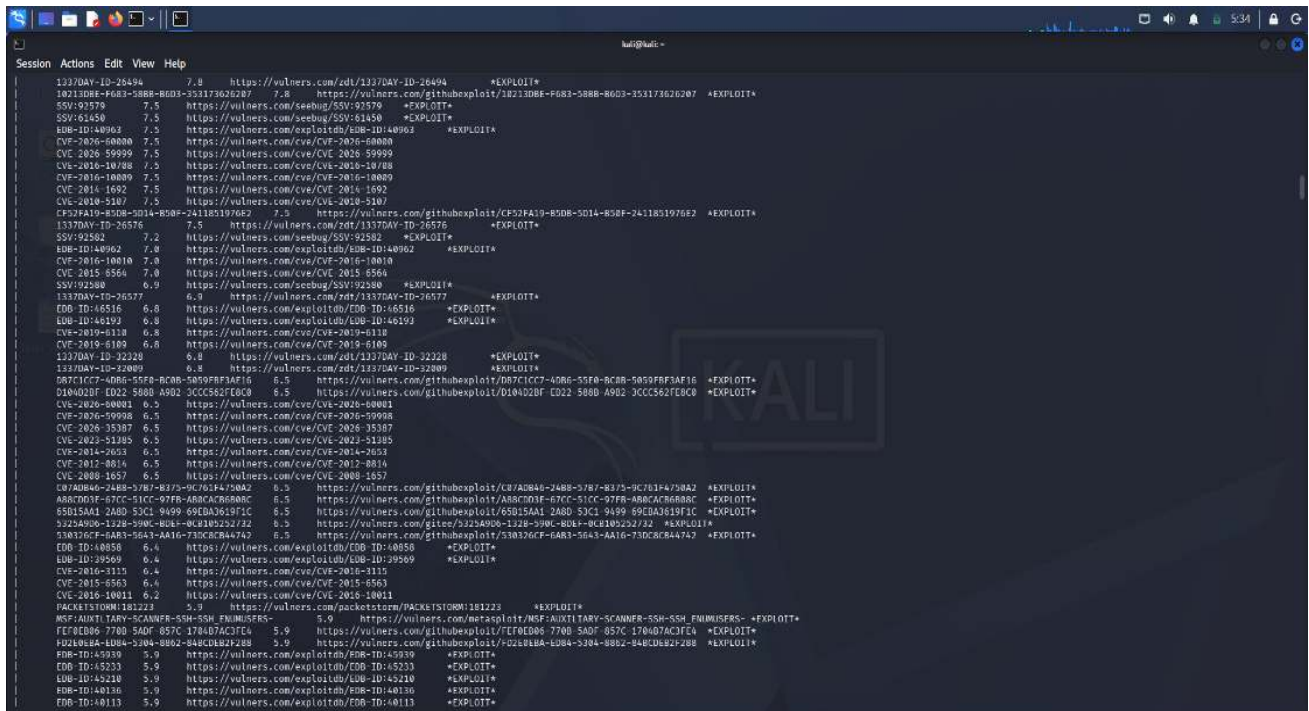
msf > nmap -sV -script vuln 192.168.56.102
[*] exec: nmap -sV --script vuln 192.168.56.102

Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-11 05:20:0400
Nmap scan report for 192.168.56.102
Host is up (0.38s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp           vsftpd 2.3.4
| vulners:
| vsftpd 2.3.4:
| PACKETSTORM-102145 10.0 https://vulners.com/packetstorm/PACKETSTORM-102145 *EXPLOIT*
| MSF-EXPLOIT-UNIX-FTP-VSFTPD_234_BACKDOOR- 10.0 https://vulners.com/metasploit/MSF-EXPLOIT-UNIX-FTP-VSFTPD_234_BACKDOOR- *EXPLOIT*
| TIA2400C-72FF-5445-8198-9C2D5322DC06 10.0 https://vulners.com/githubexploit/TIA2400C-72FF-5445-8198-9C2D5322DC06 *EXPLOIT*
| E98-1014975/ 10.0 https://vulners.com/exploitdb/E98-1014975/ *EXPLOIT*
| F06-1017491 10.0 https://vulners.com/exploitdb/F06-1017491 *EXPLOIT*
| C05791E3-8935-5632-AA81-CD7E7FA26C08 10.0 https://vulners.com/githubexploit/C05791E3-8935-5632-AA81-CD7E7FA26C08 *EXPLOIT*
| EC72873B-7726-59C3-828B-0828B8955089 10.0 https://vulners.com/githubexploit/EC72873B-7726-59C3-828B-0828B8955089 *EXPLOIT*
| F980A88B-5138-508F-8922-2087F3C04600 10.0 https://vulners.com/githubexploit/F980A88B-5138-508F-8922-2087F3C04600 *EXPLOIT*
| F356CCEC-7C81-5A77-8C7A-87DFA8D11897 10.0 https://vulners.com/githubexploit/F356CCEC-7C81-5A77-8C7A-87DFA8D11897 *EXPLOIT*
| DFC7C45F-8C41-513A-8114-88C1B0C26E11 10.0 https://vulners.com/githubexploit/DFC7C45F-8C41-513A-8114-88C1B0C26E11 *EXPLOIT*
| D8C8F348-5480-554A-8D5C-9224B40A8F9F 10.0 https://vulners.com/githubexploit/D8C8F348-5480-554A-8D5C-9224B40A8F9F *EXPLOIT*
| D8D8A85F-750F-5F08-9C69-65F8FA6977D5 10.0 https://vulners.com/githubexploit/D8D8A85F-750F-5F08-9C69-65F8FA6977D5 *EXPLOIT*
| CVE-2011-2523 10.0 https://vulners.com/cve/CVE-2011-2523
| CND-8248-48837 10.0 https://vulners.com/cnd/CND-8248-48837
| CCBF6C15-182F-53F6-A5CC-812D37F1F847 10.0 https://vulners.com/githubexploit/CCBF6C15-182F-53F6-A5CC-812D37F1F847 *EXPLOIT*
| BEF678EA-45C2-58C8-88F3-9A2E7CC2DECE 10.0 https://vulners.com/githubexploit/BEF678EA-45C2-58C8-88F3-9A2E7CC2DECE *EXPLOIT*
| A1A9B4E9-848A-546C-8222-181B21F6397 10.0 https://vulners.com/githubexploit/A1A9B4E9-848A-546C-8222-181B21F6397 *EXPLOIT*
| A180B8A8-C86F-56F7-B898-80958C840533 10.0 https://vulners.com/githubexploit/A180B8A8-C86F-56F7-B898-80958C840533 *EXPLOIT*
| A4185FAD-1A6C-56A6-97C6-1C58A1CF1E38 10.0 https://vulners.com/githubexploit/A4185FAD-1A6C-56A6-97C6-1C58A1CF1E38 *EXPLOIT*
| 998AB431-1C8F-59D8-AA48-32596997F230 10.0 https://vulners.com/githubexploit/998AB431-1C8F-59D8-AA48-32596997F230 *EXPLOIT*
| 91180FCF-48D0-502E-8527-513AC2924A49 10.0 https://vulners.com/githubexploit/91180FCF-48D0-502E-8527-513AC2924A49 *EXPLOIT*
| 885D885A-A37F-5A73-5F9A-FF3F78D0F580 10.0 https://vulners.com/githubexploit/885D885A-A37F-5A73-5F9A-FF3F78D0F580 *EXPLOIT*
| 88854709-11C3-5D43-86E4-387A3D4ACCA8 10.0 https://vulners.com/githubexploit/88854709-11C3-5D43-86E4-387A3D4ACCA8 *EXPLOIT*
| 817C0B1-87CA-5948-B992-9CC6433A3A33 10.0 https://vulners.com/githubexploit/817C0B1-87CA-5948-B992-9CC6433A3A33 *EXPLOIT*
| 74AC1193-6858-5FA3-8845-FCB878736833 10.0 https://vulners.com/githubexploit/74AC1193-6858-5FA3-8845-FCB878736833 *EXPLOIT*
| 71C10AC4-8A80-556B-AD36-E0C2056780E7 10.0 https://vulners.com/githubexploit/71C10AC4-8A80-556B-AD36-E0C2056780E7 *EXPLOIT*
| 84B6E781-0C80-5C22-8B45-C12AF7CA4788 10.0 https://vulners.com/githubexploit/84B6E781-0C80-5C22-8B45-C12AF7CA4788 *EXPLOIT*
| 63AC0A7F-C241-5881-98F6-ABA8448D0278 10.0 https://vulners.com/githubexploit/63AC0A7F-C241-5881-98F6-ABA8448D0278 *EXPLOIT*
| 5FA8CCE2-770F-505A-851A-BAE8B76A5089 10.0 https://vulners.com/githubexploit/5FA8CCE2-770F-505A-851A-BAE8B76A5089 *EXPLOIT*
| 5F0CAF4F-8FE6-5480-9162-3EE808989868 10.0 https://vulners.com/githubexploit/5F0CAF4F-8FE6-5480-9162-3EE808989868 *EXPLOIT*
| 59B8FDCD-5027-5C82-8C59-FC7623D081D8 10.0 https://vulners.com/githubexploit/59B8FDCD-5027-5C82-8C59-FC7623D081D8 *EXPLOIT*
| 578AA185-2908-5F48-8A48-3C81D25F8687 10.0 https://vulners.com/githubexploit/578AA185-2908-5F48-8A48-3C81D25F8687 *EXPLOIT*
| 5062D0A4-437D-5156-8197-AD027A9A660C 10.0 https://vulners.com/githubexploit/5062D0A4-437D-5156-8197-AD027A9A660C *EXPLOIT*
| 50589896-73C4-2097-81C4-34D0D091D444 10.0 https://vulners.com/githubexploit/50589896-73C4-2097-81C4-34D0D091D444 *EXPLOIT*
| 508C71C1-8183-56FA-8198-1525F11832F1 10.0 https://vulners.com/githubexploit/508C71C1-8183-56FA-8198-1525F11832F1 *EXPLOIT*
| A061A1A1-75C7-5738-8C5D-02C1F56532F 10.0 https://vulners.com/githubexploit/A061A1A1-75C7-5738-8C5D-02C1F56532F *EXPLOIT*
| 208FA416-98C2-53C2-951E-88A6C02298A8 10.0 https://vulners.com/githubexploit/208FA416-98C2-53C2-951E-88A6C02298A8 *EXPLOIT*
| 344AF37C-35F5-5A78-838A-889587233D08 10.0 https://vulners.com/githubexploit/344AF37C-35F5-5A78-838A-889587233D08 *EXPLOIT*
| 322D01ED-0331-573E-9A4C-5868FC2895F9 10.0 https://vulners.com/githubexploit/322D01ED-0331-573E-9A4C-5868FC2895F9 *EXPLOIT*
```

3. Vulnerability Findings

The assessment identified multiple intentionally vulnerable services. The screenshots below preserve the observed evidence used to support the findings.

04 Vulnerability Finding 1



```
Session Actions Edit View Help
1337DAY-ID-26494 7.8 https://vulners.com/cve/1337DAY-ID-26494 *EXPLOIT*
1821308E-F683-588B-86D3-353173626287 7.8 https://vulners.com/githubexploit/1821308E-F683-588B-86D3-353173626287 *EXPLOIT*
SSV:92572 7.5 https://vulners.com/seebug/SSV:92572 *EXPLOIT*
SSV:61450 7.5 https://vulners.com/seebug/SSV:61450 *EXPLOIT*
E08-ID:40963 7.5 https://vulners.com/exploitdb/E08-ID:40963 *EXPLOIT*
CVE-2026-68000 7.5 https://vulners.com/cve/CVE-2026-68000
CVE-2026-59999 7.5 https://vulners.com/cve/CVE-2026-59999
CVE-2016-18788 7.5 https://vulners.com/cve/CVE-2016-18788
CVE-2016-10009 7.5 https://vulners.com/cve/CVE-2016-10009
CVE-2014-1692 7.5 https://vulners.com/cve/CVE-2014-1692
CVE-2018-5187 7.5 https://vulners.com/cve/CVE-2018-5187
CPS2FA19-85D8-5014-85D8-241185197682 7.5 https://vulners.com/githubexploit/CPS2FA19-85D8-5014-85D8-241185197682 *EXPLOIT*
1337DAY-ID-26576 7.5 https://vulners.com/cve/1337DAY-ID-26576 *EXPLOIT*
SSV:92582 7.2 https://vulners.com/seebug/SSV:92582 *EXPLOIT*
E08-ID:40962 7.0 https://vulners.com/exploitdb/E08-ID:40962 *EXPLOIT*
CVE-2016-10010 7.0 https://vulners.com/cve/CVE-2016-10010
CVE-2015-6564 7.0 https://vulners.com/cve/CVE-2015-6564
SSV:92580 6.9 https://vulners.com/seebug/SSV:92580 *EXPLOIT*
1337DAY-ID-26577 6.9 https://vulners.com/cve/1337DAY-ID-26577 *EXPLOIT*
CDB-ID:46516 6.8 https://vulners.com/exploitdb/CDB-ID:46516 *EXPLOIT*
E08-ID:46192 6.8 https://vulners.com/exploitdb/E08-ID:46192 *EXPLOIT*
CVE-2019-6118 6.8 https://vulners.com/cve/CVE-2019-6118
CVE-2019-6109 6.8 https://vulners.com/cve/CVE-2019-6109
1337DAY-ID-32228 6.8 https://vulners.com/cve/1337DAY-ID-32228 *EXPLOIT*
1337DAY-ID-32009 6.8 https://vulners.com/cve/1337DAY-ID-32009 *EXPLOIT*
D87C1CCT-4086-50F8-8C8B-5809F8F5A16 6.5 https://vulners.com/githubexploit/D87C1CCT-4086-50F8-8C8B-5809F8F5A16 *EXPLOIT*
D1A0D20F C022 50B0 A902 3CC562FEC8 6.5 https://vulners.com/githubexploit/D1A0D20F C022 50B0 A902 3CC562FEC8 *EXPLOIT*
CVE-2026-50001 6.5 https://vulners.com/cve/CVE-2026-50001
CVE-2026-59998 6.5 https://vulners.com/cve/CVE-2026-59998
CVE-2026-35387 6.5 https://vulners.com/cve/CVE-2026-35387
CVE-2022-51185 6.5 https://vulners.com/cve/CVE-2022-51185
CVE-2014-2653 6.5 https://vulners.com/cve/CVE-2014-2653
CVE-2012-8814 6.5 https://vulners.com/cve/CVE-2012-8814
CVE-2008-1657 6.5 https://vulners.com/cve/CVE-2008-1657
C87A0B46-2488-5787-8375-9C761F4798A2 6.5 https://vulners.com/githubexploit/C87A0B46-2488-5787-8375-9C761F4798A2 *EXPLOIT*
ARRCD03F-67CC-51CC-97F8-ARRACR6800AC 6.5 https://vulners.com/githubexploit/ARRCD03F-67CC-51CC-97F8-ARRACR6800AC *EXPLOIT*
65015AA1 2A80 53C1 9499 69EDA3619F1C 6.5 https://vulners.com/githubexploit/65015AA1 2A80 53C1 9499 69EDA3619F1C *EXPLOIT*
32A5A906-1228-598C-B0E5-9C818252722 6.5 https://vulners.com/cve/32A5A906-1228-598C-B0E5-9C818252722 *EXPLOIT*
53032CF-6A83-5643-AA16-73DC8C844742 6.5 https://vulners.com/githubexploit/53032CF-6A83-5643-AA16-73DC8C844742 *EXPLOIT*
CDB-ID:40858 6.4 https://vulners.com/exploitdb/CDB-ID:40858 *EXPLOIT*
E08-ID:39569 6.4 https://vulners.com/exploitdb/E08-ID:39569 *EXPLOIT*
CVE-2016-1113 6.4 https://vulners.com/cve/CVE-2016-1113
CVE-2015-6563 6.4 https://vulners.com/cve/CVE-2015-6563
CVE-2016-10011 6.2 https://vulners.com/cve/CVE-2016-10011
PACKETSTORM181223 5.9 https://vulners.com/packetstorm/PACKETSTORM181223 *EXPLOIT*
MSF-AUXILIARY-SCANNER-SSH-SM-FWMINISERS- 5.9 https://vulners.com/githubexploit/MSF-AUXILIARY-SCANNER-SSH-SM-FWMINISERS- *EXPLOIT*
FEBED06 7780 5ADF 857C 1784D7AC37C4 5.9 https://vulners.com/githubexploit/FEBED06 7780 5ADF 857C 1784D7AC37C4 *EXPLOIT*
H2408BA-E084-5304-8862-84CD6B2F288 5.9 https://vulners.com/githubexploit/H2408BA-E084-5304-8862-84CD6B2F288 *EXPLOIT*
E08-ID:45939 5.9 https://vulners.com/exploitdb/E08-ID:45939 *EXPLOIT*
CDB-ID:45233 5.9 https://vulners.com/exploitdb/CDB-ID:45233 *EXPLOIT*
E08-ID:45210 5.9 https://vulners.com/exploitdb/E08-ID:45210 *EXPLOIT*
E08-ID:40136 5.9 https://vulners.com/exploitdb/E08-ID:40136 *EXPLOIT*
E08-ID:40113 5.9 https://vulners.com/exploitdb/E08-ID:40113 *EXPLOIT*
```

05 Vulnerability Finding 2

```
kali@kali: ~  
Session Actions Edit View Help  
|_ 1337DAY-ID-27896 0.0 https://vulners.com/cve/1337DAY-ID-27896 *EXPLOIT*  
80/tcp open http Apache httpd 2.2.8 ((Ubuntu) DAV/2)  
| http-squid:  
| /ikiwiki/: TwikiWiki  
| /test/: Test page  
| /papiinfo.php: Possible information file  
| /phpMyAdmin/: phpMyAdmin  
| /doc/: Potentially interesting directory w/ listing on 'apache/2.2.8 (ubuntu) dav/2'  
| /icons/: Potentially interesting folder w/ directory listing  
| /index/: Potentially interesting folder  
|_ http-csrf:  
| Spidering limited to: maxdepth=3; maxpagecount=20; withinhost=192.168.56.102  
| Found the following possible CSRF vulnerabilities:  
|  
| Path: http://192.168.56.102:80/dvwa/  
| Form id:  
| Form action: login.php  
|  
| Path: http://192.168.56.102:80/twiki/TWikiDocumentation.html  
| Form id:  
| Form action: http://Twiki.org/cgi-bin/passwd/TWiki/WebHome  
|  
| Path: http://192.168.56.102:80/twiki/TWikiDocumentation.html  
| Form id:  
| Form action: http://Twiki.org/cgi-bin/passwd/Main/WebHome  
|  
| Path: http://192.168.56.102:80/twiki/TWikiDocumentation.html  
| Form id:  
| Form action: http://Twiki.org/cgi-bin/edit/TWiki/  
|  
| Path: http://192.168.56.102:80/twiki/TWikiDocumentation.html  
| Form id:  
| Form action: http://Twiki.org/cgi-bin/view/TWiki/TWikiSkins  
|  
| Path: http://192.168.56.102:80/twiki/TWikiDocumentation.html  
| Form id:  
| Form action: http://Twiki.org/cgi-bin/manage/TWiki/ManagingWebs  
|  
| Path: http://192.168.56.102:80/dvwa/login.php  
| Form id:  
| Form action: login.php  
|_ http-fileupload-exploiter:  
|  
| Couldn't find a file-type field.  
|_ http vuln cve2017-1001000: ERROR: Script execution failed (use -d to debug)  
|_ http-stored-xss: Couldn't find any stored XSS vulnerabilities.  
|_ vulners:  
|_ cpe:/a:apache:http_server:2.2.8:  
|_ SSV:09341 10.0 https://vulners.com/seebug/SSV:09341 *EXPLOIT*  
|_ SSV:19202 10.0 https://vulners.com/seebug/SSV:19202 *EXPLOIT*  
|_ SSV:19226 10.0 https://vulners.com/seebug/SSV:19226 *EXPLOIT*  
|_ SSV:11999 10.0 https://vulners.com/seebug/SSV:11999 *EXPLOIT*  
|_ PACKETSTORM:86964 10.0 https://vulners.com/packetstorm/PACKETSTORM:86964 *EXPLOIT*
```

06 Vulnerability Finding 3

```
Session Actions Edit View Help
| CVE-2007-5925 4.0 https://vulners.com/cve/CVE-2007-5925
| CVE-2010-1626 3.6 https://vulners.com/cve/CVE-2010-1626
| SSL/TLS MITM vulnerability (CCS Injection)
| STATE: VULNERABLE
| Risk factor: High
| OpenSSL before 0.9.8za, 1.0.0 before 1.0.0m, and 1.0.1 before 1.0.1h
| does not properly restrict processing of ChangeCipherSpec messages,
| which allows man-in-the-middle attackers to trigger use of a zero
| length master key in certain OpenSSL-to-OpenSSL communications, and
| consequently hijack sessions or obtain sensitive information, via
| a crafted TLS handshake, aka the "CCS Injection" vulnerability.
|
| References:
| http://www.cvedetails.com/cve/2014-0224
| http://www.openssl.org/news/secadv_20140805.txt
| https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2014-0224
|
| ssl-poodle:
| VULNERABLE:
| SSL POODLE information leak
| STATE: VULNERABLE
| IDs: BID:78574 CVE:CVE-2014-3566
| The SSL protocol 3.0, as used in OpenSSL through 1.0.1i and other
| products, uses nondeterministic CBC padding, which makes it easier
| for man-in-the-middle attackers to obtain cleartext data via a
| padding-oracle attack, aka the "POODLE" issue.
| Disclosure date: 2014-10-14
| Check results:
| TLS_RSA_WITH_AES_128_CBC_SHA
| References:
| https://www.openssl.org/~bodo/ssl-poodle.pdf
| https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2014-3566
| https://www.imperialviolet.org/2014/10/14/poodle.html
| https://www.securityfocus.com/bid/78574
|
| vulners:
| cpe:/a:postgresql:postgresql:8.21
| SSV:60718 10.0 https://vulners.com/seebug/SSV:60718 *EXPLOIT*
| CVE-2013-1903 10.0 https://vulners.com/cve/CVE-2013-1903
| CVE-2013-1902 10.0 https://vulners.com/cve/CVE-2013-1902
| PACKETSTORM:218218 9.9 https://vulners.com/packetstorm/PACKETSTORM:218218 *EXPLOIT*
| POSTGRESQL-CVE-2010-10711 9.8 https://vulners.com/postgresql/POSTGRESQL-CVE-2010-10711
| POSTGRESQL-CVE-2010-16850 9.8 https://vulners.com/postgresql/POSTGRESQL-CVE-2010-16850
| POSTGRESQL-CVE-2017-7346 9.8 https://vulners.com/postgresql/POSTGRESQL-CVE-2017-7346
| POSTGRESQL-CVE-2015-3166 9.8 https://vulners.com/postgresql/POSTGRESQL-CVE-2015-3166
| POSTGRESQL-CVE-2015-0244 9.8 https://vulners.com/postgresql/POSTGRESQL-CVE-2015-0244
| PACKETSTORM:189310 9.8 https://vulners.com/packetstorm/PACKETSTORM:189310 *EXPLOIT*
| CVE-2010-10711 9.8 https://vulners.com/cve/CVE-2010-10711
| CVE-2015-3166 9.8 https://vulners.com/cve/CVE-2015-3166
| CVE-2015-0244 9.8 https://vulners.com/cve/CVE-2015-0244
| CNVD-2020-02196 9.8 https://vulners.com/cnvd/CNVD-2020-02196
| CNVD-2017-26577 9.8 https://vulners.com/cnvd/CNVD-2017-26577
```

07 Vulnerability Finding 4

```

CVS-2014-0060 4.0 https://vulners.com/cve/CVE-2014-0060
CVE-2012-2655 4.0 https://vulners.com/cve/CVE-2012-2655
CVE-2009-3229 4.0 https://vulners.com/cve/CVE-2009-3229
POSTGRESQ-CVE-2025-4669 3.0 https://vulners.com/postgresql/POSTGRESQ-CVE-2025-4669
POSTGRESQ-CVE-2026-14673 3.8 https://vulners.com/postgresql/POSTGRESQ-CVE-2026-14673
POSTGRESQ-CVE-2026-14673 3.8 https://vulners.com/postgresql/POSTGRESQ-CVE-2026-14673
POSTGRESQ-CVE-2024-19977 3.7 https://vulners.com/postgresql/POSTGRESQ-CVE-2024-19977
POSTGRESQ-CVE-2022-41882 3.7 https://vulners.com/postgresql/POSTGRESQ-CVE-2022-41882
SSV19322 3.5 https://vulners.com/ssh/ssv19322 *EXPLOIT*
POSTGRESQ-CVE-2019-10209 3.5 https://vulners.com/postgresql/POSTGRESQ-CVE-2019-10209
PACKETSTORM:127092 3.5 https://vulners.com/packetstorm/PACKETSTORM:127092 *EXPLOIT*
FBI-13379 3.5 https://vulners.com/exploit/FBI-13379 *EXPLOIT*
CVE-2010-8733 3.5 https://vulners.com/cve/CVE-2010-8733
POSTGRESQ-CVE-2025-8713 3.1 https://vulners.com/postgresql/POSTGRESQ-CVE-2025-8713
POSTGRESQ-CVE-2025-12817 3.1 https://vulners.com/postgresql/POSTGRESQ-CVE-2025-12817
ssl-params:
VULNERABLE:
Diffie-Hellman Key Exchange Insufficient Group Strength
State: VULNERABLE
Transport Layer Security (TLS) services that use Diffie-Hellman groups
of insufficient strength, especially those using one of a few commonly
shared groups, may be susceptible to passive eavendropping attacks.
Check results:
WEAK ON GROUP 1
Cipher Suite: TLS_DHE_RSA_WITH_3DES_EDE_CBC_SHA
Modulus Type: Safe primes
Modulus Source: Unknown/Custom-generated
Modulus Length: 1024
Generator Length: 8
Public Key Length: 1024
References:
https://weakdh.org
5000/tcp open vnc VNC (protocol 3.3)
8000/tcp open x11 (access denied)
6666/tcp open irc UnrealIRCd
|_irc-unrealircd-backdoor: looks like trojaned version of unrealircd. See http://seclists.org/fulldisclosure/2010/jun/277
8009/tcp open sjs13 Apache Jserv (Protocol v1.3)
8180/tcp open http Apache/loncat/Coyote JSP engine 1.1
|_http-forward-xxx: Couldn't find any stored XSS vulnerabilities.
|_http-server-header: Apache-Coyote/1.1
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-database-xxx: Couldn't find any DOM based XSS.
MAC Address: 08:00:27:82:32:8F (Oracle VM VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, lrc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
_smb-vuln-ms08-051: False
_smb-vuln-regsvcs-dos: UNKNOWN: Script execution failed (use -d to debug)
_smb-vuln-ms10-061: False

Service detection performed. Please report any incorrect results at https://map.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 304.19 seconds
msf >

```

4. VSFTPD 2.3.4 — Vulnerability Validation

TCP/21 was identified as running vsftpd 2.3.4. Metasploit was opened and the VSFTPD 2.3.4 backdoor module was selected. Module information identifies the historical backdoor and CVE-2011-2523. The module options show the target host and FTP port as required parameters.

Impact: A vulnerable/backdoored FTP service can enable unauthorized command execution and compromise of the target.

Mitigation: Remove or upgrade the vulnerable service, restrict FTP exposure, prefer secure file transfer, and monitor logs.

Metasploit VSFTPD 2.3.4 module

[illegible]

Module information and CVE reference

```
kali@kali: ~$ msf5 > info 0
# Name Disclosure Date Rank Check Description
0 exploit/unix/ftp/vsftpd_23k_backdoor 2011-07-03 excellent Yes vsftpd 2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/ftp/vsftpd_23k_backdoor

msf5 > info 0
Name: VSFTPD 2.3.4 Backdoor Command Execution
Module: exploit/unix/ftp/vsftpd_23k_backdoor
Platform: Unix, Linux
Arch: cmd
Privileged: Yes
License: Metasploit Framework License (BSD)
Rank: Excellent
Disclosed: 2011-07-03

Provided by:
hdm <hdm@hdm.io>
MC <mc@metasploit.com>
gatemik

Module side effects:
unknown side effects

Module stability:
unknown stability

Module reliability:
unknown reliability

Available targets:
id Name
-- --
-> 0 Linux/Unix Command

Check supported:
Yes

Basic options:
Name Current Setting Required Description
--
RHOSTS 21 yes The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT 21 yes The target port (TCP)

Payload information:
Space: 2000
Avoid: 0 characters

Description:
This module exploits a malicious backdoor that was added to the VSFTPD download archive. This backdoor was introduced into the vsftpd-2.3.4.tar.gz archive between
```

Module configuration options

```
kali@kali:~$ msf > use 0
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf > exploit(0x4/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):



| Name   | Current Setting | Required | Description                                                                                            |
|--------|-----------------|----------|--------------------------------------------------------------------------------------------------------|
| RHOSTS | 21              | yes      | The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html |
| RPORT  | 21              | yes      | The target port (TCP)                                                                                  |



Payload options (cmd/linux/http/x86/meterpreter_reverse_tcp):



| Name           | Current Setting | Required | Description                                                                                                                                                                                                             |
|----------------|-----------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FETCH_COMMAND  | CURL            | yes      | Command to fetch payload (Accepted: CURL, FTP, TFTP, NNTP, WGET)                                                                                                                                                        |
| FETCH_DELETE   | false           | yes      | Attempt to delete the binary after execution                                                                                                                                                                            |
| FETCH_FILELESS | none            | yes      | Attempt to run payload without touching disk by using anonymous handles, requires Linux >3.17 (for Python variant also Python >3.8, tested shells are sh, bash, zsh) (Accepted: none, python 3.8+, shell-search, shell) |
| FETCH_SRVHOST  |                 | no       | Local IP to use for serving payload                                                                                                                                                                                     |
| FETCH_SRVPORT  | 8888            | yes      | Local port to use for serving payload                                                                                                                                                                                   |
| FETCH_URI_PATH |                 | no       | Local URI to use for serving payload                                                                                                                                                                                    |
| LHOST          |                 | yes      | The listen address (an interface may be specified)                                                                                                                                                                      |
| LPORT          | 4444            | yes      | The listen port                                                                                                                                                                                                         |



When FETCH_COMMAND is one of CURL,Git,WGET:



| Name       | Current Setting | Required | Description                                                                            |
|------------|-----------------|----------|----------------------------------------------------------------------------------------|
| FETCH_PIPE | false           | yes      | Host both the binary payload and the command so it can be piped directly to the shell. |



When FETCH_FILELESS is none:



| Name               | Current Setting | Required | Description                                                                         |
|--------------------|-----------------|----------|-------------------------------------------------------------------------------------|
| FETCH_FILENAME     | hdon3qHafck     | no       | Name to use on remote system when storing payload; cannot contain spaces or slashes |
| FETCH_WRITABLE_DIR | ./              | yes      | Remote writable dir to store payload; cannot contain spaces                         |



Exploit target:



| ID | Name               |
|----|--------------------|
| 0  | Linux/Unix Command |



View the full module info with the info, or info -d command.

msf > exploit(0x4/ftp/vsftpd_234_backdoor) > 
```

5. SSH Security Assessment

SSH was observed on TCP/22. Algorithm enumeration showed legacy options, including SHA-1 based key-exchange algorithms, SSH-RSA/SSH-DSS host-key algorithms, older CBC/ARCFOUR-style encryption options, and legacy MACs. These observations support a recommendation to modernize the SSH configuration.

Mitigation: Disable deprecated algorithms, use a current OpenSSH release, restrict SSH to trusted networks, and use strong authentication controls.

SSH security assessment

```

Session Actions Edit View Help
[..._ssh-vuln-m218-054: false
[..._ssh-vuln-recv-dos: ERROR: Script execution failed (use -d to debug)
[..._ssh-vuln-m218-061: false

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 364.19 seconds
[...$ nmap -p 22 -script ssh2-enum-algos 192.168.56.102
[...$ exec: nmap -p 22 --script ssh2-enum-algos 192.168.56.102

Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-11 05:39 -0400
Nmap scan report for 192.168.56.102
Host is up (0.00002s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
[...ssh-enum-algos:
[...  key_algorithms: (1)
[...    diffie-hellman-group-exchange-sha256
[...    diffie-hellman-group-exchange-sha1
[...    diffie-hellman-group1-sha1
[...    diffie-hellman-group1-sha1
[...  server_host_key_algorithms: (2)
[...    ssh-rsa
[...    ssh-dss
[...  encryption_algorithms: (13)
[...    aes128-ctr
[...    3des-ctr
[...    blowfish-ctr
[...    cast128-ctr
[...    arcfour128
[...    arcfour256
[...    arcfour
[...    aes192-ctr
[...    aes256-ctr
[...    rijndael-ctr@lysator.lia.se
[...    aes128-ctr
[...    aes192-ctr
[...    aes256-ctr
[...  mac_algorithms: (7)
[...    hmac-md5
[...    hmac-sha1
[...    umac-64@openssh.com
[...    hmac-ripemd160
[...    hmac-ripemd160@openssh.com
[...    hmac-sha1-96
[...    hmac-md5-96
[...  compression_algorithms: (2)
[...    none
[...    zlib@openssh.com
MAC Address: 08:00:27:02:32:0E (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.73 seconds
[...$
```

SSH algorithm enumeration

```

kali@kali:~$ sudo nmap -p 22 -sV 192.168.56.102
Nmap scan report for 192.168.56.102
Host is up (0.812s latency).

```

View the full module info with the `info`, or `info -d` command.

```

kali@kali:~$ nmap --script ssh2-enum-algos 192.168.56.102
exec: nmap -p 22 --script ssh2-enum-algos 192.168.56.102

```

Starting Nmap 7.99 (<https://nmap.org>) at 2026-09-11 06:45 -0400

```

Nmap scan report for 192.168.56.102
Host is up (0.812s latency).

```

PORT STATE SERVICE

```

22/tcp open  ssh

```

ssh2-enum-algos:

```

  Ker algorithms: (1)
    diffie-hellman-group-exchange-sha256
    diffie-hellman-group-exchange-sha1
    diffie-hellman-group1-sha1
    diffie-hellman-group1-sha1
  server host key algorithms: (2)
    ssh-rsa
    ssh-dss
  encryption algorithms: (13)
    aes128-ctr
    aes128-cbc
    3des-ctr
    blowfish-ctr
    cast128-ctr
    arcfour128
    arcfour256
    arcfour
    aes192-ctr
    aes256-ctr
    rijndael-cbc@lysator.liu.se
    aes128-ctr
    aes192-ctr
    aes256-ctr
  mac algorithms: (7)
    hmac-md5
    hmac-sha1
    umac-64@openssh.com
    hmac-ripemd160
    hmac-ripemd160@openssh.com
    hmac-sha1-96
    hmac-md5-96
  compression algorithms: (2)
    none
    zlib@openssh.com

```

MAC Address: 08:00:27:62:32:0E (Oracle VM VirtualBox virtual NIC)

```

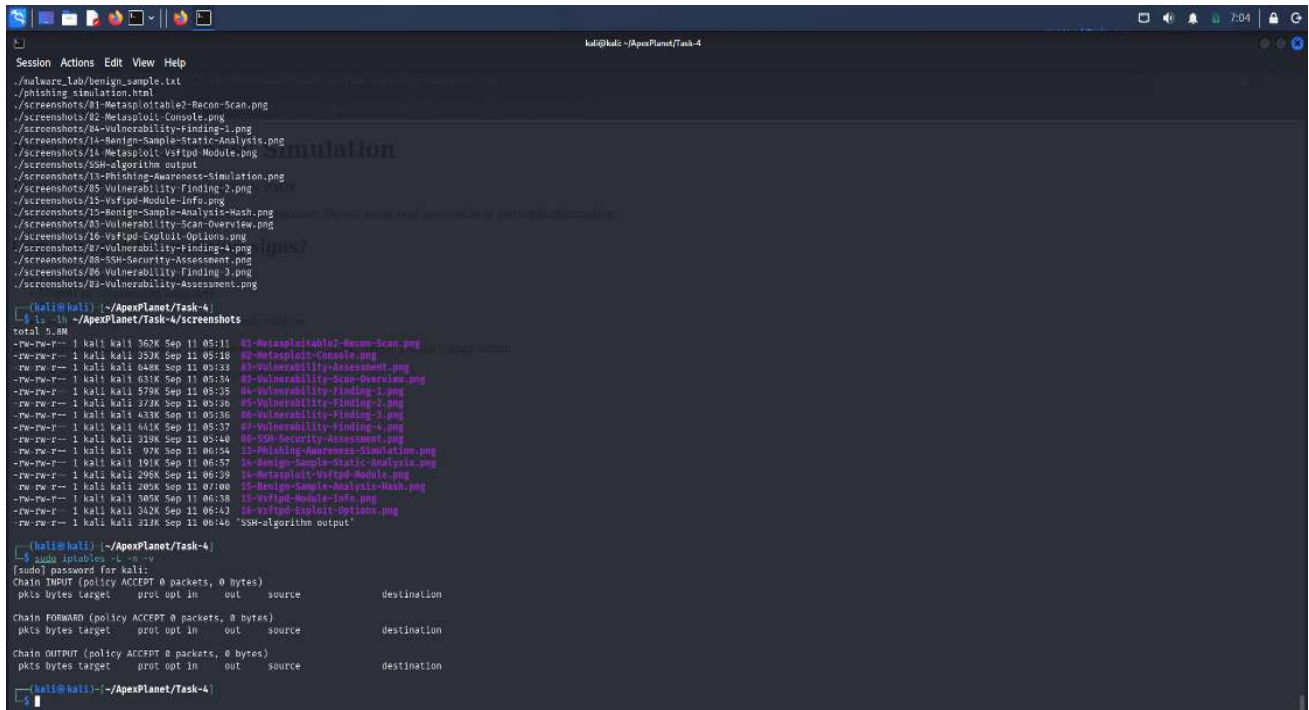
Nmap done: 1 IP address (1 host up) scanned in 0.77 seconds
kali@kali:~$ sudo nmap -p 22 -sV 192.168.56.102

```

6. System Hardening Assessment

The lab included evidence collection for firewall state, listening services, and patch/upgrade status. The recommended hardening actions are to restrict inbound traffic, disable unnecessary services, and apply security updates.

Firewall state



```
kali@kali: ~/ApexPlanet/Task-4
Session Actions Edit View Help
./vulnware_lab/benign_sample.txt
./phishing_simulation.html
./screenshots/01-Metasploit-Beacon-Scan.png
./screenshots/02-Metasploit-Console.png
./screenshots/04-Vulnerability-Finding-1.png
./screenshots/14-Benign-Sample-Static-Analysis.png
./screenshots/14-Metasploit-Vsftpd-Module.png
./screenshots/259M-algorithm_output
./screenshots/15-Pushing-Awareness-Simulation.png
./screenshots/05-Vulnerability-Finding-2.png
./screenshots/15-Vsftpd-Module-Info.png
./screenshots/15-Benign-Sample-Analysis-Hash.png
./screenshots/03-Vulnerability-Scan-Overview.png
./screenshots/16-Vsftpd-Exploit-Options.png
./screenshots/07-Vulnerability-Finding-4.png
./screenshots/08-SSH-Security-Assessment.png
./screenshots/06-Vulnerability-Finding-3.png
./screenshots/03-Vulnerability-Assessment.png

kali@kali: ~/ApexPlanet/Task-4
ls -lh ~/ApexPlanet/Task-4/screenshots
total 5.8M
-rw-rw-r-- 1 kali kali 362K Sep 11 05:13 03-Metasploit-Beacon-Scan.png
-rw-rw-r-- 1 kali kali 252K Sep 11 05:10 02-Metasploit-Console.png
-rw-rw-r-- 1 kali kali 648K Sep 11 02:13 03-Vulnerability-Assessment.png
-rw-rw-r-- 1 kali kali 631K Sep 11 05:34 03-Vulnerability-Scan-Overview.png
-rw-rw-r-- 1 kali kali 579K Sep 11 05:35 04-Vulnerability-Finding-1.png
-rw-rw-r-- 1 kali kali 372K Sep 11 02:30 05-Vulnerability-Finding-2.png
-rw-rw-r-- 1 kali kali 433K Sep 11 05:36 06-Vulnerability-Finding-3.png
-rw-rw-r-- 1 kali kali 441K Sep 11 05:37 07-Vulnerability-Finding-4.png
-rw-rw-r-- 1 kali kali 219K Sep 11 05:40 08-SSH-Security-Assessment.png
-rw-rw-r-- 1 kali kali 97K Sep 11 06:54 13-Pushing-Awareness-Simulation.png
-rw-rw-r-- 1 kali kali 191K Sep 11 06:57 14-Benign-Sample-Static-Analysis.png
-rw-rw-r-- 1 kali kali 296K Sep 11 06:39 14-Metasploit-Vsftpd-Module.png
-rw-rw-r-- 1 kali kali 200K Sep 11 07:00 15-Benign-Sample-Analysis-Hash.png
-rw-rw-r-- 1 kali kali 300K Sep 11 06:30 15-Vsftpd-Module-Info.png
-rw-rw-r-- 1 kali kali 342K Sep 11 06:43 16-Vsftpd-Exploit-Options.png
-rw-rw-r-- 1 kali kali 212K Sep 11 06:46 'SSH-algorithm_output'

kali@kali: ~/ApexPlanet/Task-4
sudo iptables -L -n -v
[sudo] password for kali:
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in     out     source            destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in     out     source            destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target      prot opt in     out     source            destination
```

Listening services

```
kali@kali: ~/ApexPlanet/Task-4
Session Actions Edit View Help
./screenshots/SSH-algorithm output
./screenshots/13-Phishing-Awareness-Simulation.png
./screenshots/03-Vulnerability-Finding-2.png
./screenshots/15-VsFipd-Module-Info.png
./screenshots/15-Benign-Sample-Analysis-Hash.png
./screenshots/03-Vulnerability-Scan-Overview.png
./screenshots/15-VsFipd-Exploit-Options.png
./screenshots/03-Vulnerability-Finding-4.png
./screenshots/08-SSM-Security-Assessment.png
./screenshots/06-Vulnerability-Finding-3.png
./screenshots/03-Vulnerability-Assessment.png
[ali@kali] ~/ApexPlanet/Task-4
ls -lh ~/ApexPlanet/Task-4/screenshots
total 9.0M
-rw-rw-r-- 1 kali kali 367K Sep 11 05:11 01-Metasploitables-Recm-Scan.png
-rw-rw-r-- 1 kali kali 352K Sep 11 05:10 02-Metasploit-Console.png
-rw-rw-r-- 1 kali kali 048K Sep 11 05:12 03-Vulnerability-Assessment.png
-rw-rw-r-- 1 kali kali 031K Sep 11 05:14 04-Vulnerability-Scan-Overview.png
-rw-rw-r-- 1 kali kali 579K Sep 11 05:15 04-Vulnerability-Finding-1.png
-rw-rw-r-- 1 kali kali 272K Sep 11 05:16 05-Vulnerability-Finding-2.png
-rw-rw-r-- 1 kali kali 433K Sep 11 05:16 06-Vulnerability-Finding-3.png
-rw-rw-r-- 1 kali kali 441K Sep 11 05:17 07-Vulnerability-Finding-4.png
-rw-rw-r-- 1 kali kali 219K Sep 11 05:10 08-SSM-Security-Assessment.png
-rw-rw-r-- 1 kali kali 97K Sep 11 06:14 13-Phishing-Awareness-Simulation.png
-rw-rw-r-- 1 kali kali 191K Sep 11 06:17 14-Benign-Sample-Static-Analysis.png
-rw-rw-r-- 1 kali kali 290K Sep 11 06:19 14-Metasploit-VsFipd-Module.png
-rw-rw-r-- 1 kali kali 200K Sep 11 07:00 15-Benign-Sample-Analysis-Hash.png
-rw-rw-r-- 1 kali kali 180K Sep 11 06:18 15-VsFipd-Module-Info.png
-rw-rw-r-- 1 kali kali 342K Sep 11 06:13 16-VsFipd-Exploit-Options.png
-rw-rw-r-- 1 kali kali 213K Sep 11 06:10 'SSH-algorithm output'
[ali@kali] ~/ApexPlanet/Task-4
sudo iptables -t -n -v
[sudo] password for kali:
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target prot opt in out source destination
Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target prot opt in out source destination
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
pkts bytes target prot opt in out source destination
[ali@kali] ~/ApexPlanet/Task-4
netstat -tlnp
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address Foreign Address State PID/Program name
```

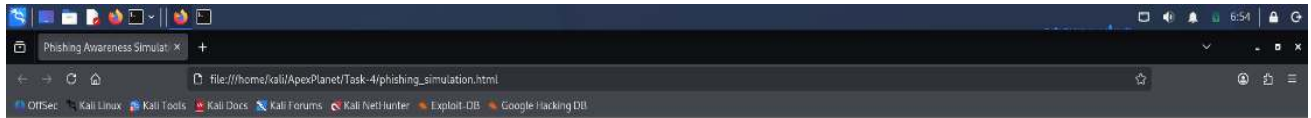
Patch assessment

```
Session Actions Edit View Help
Conf ruby-linfo (2.0.6-2 kali-rolling [all])
Conf ruby-lisa (1.15.2-1 kali-rolling [all])
Conf ruby-connection-pool (3.0.2-2 kali-rolling [all])
Conf ruby-minitest (5.0.6-2 kali-rolling [all])
Conf ruby-activesupport (2.7.2.2-2dfsg-2 kali-rolling [all])
Conf ruby-mtn-portfile (2.8.9-7 kali-rolling [all])
Conf ruby-rokgiri (1.19.4-dfsg-1 kali-rolling [amd64])
Conf ruby-opt-parse-validator (1.18.1-0kali2 kali-rolling [all])
Conf ruby-yajl (1.4.3-7 kali-rolling [amd64])
Conf ruby-webrick (1.9.2-2 kali-rolling [all])
Conf ruby-enum-scanner (0.25.0-0kali4 kali-rolling [all])
Conf ruby-public-suffix (7.0.5-1 kali-rolling [all])
Conf ruby-addressable (2.9.0-1 kali-rolling [all])
Conf ruby-ipaddress (0.8.2-6 kali-rolling [all])
Conf ruby-levenshtein (0.2.2-4 kali-rolling [amd64])
Conf ruby-little-plugger (1.1.4-3 kali-rolling [all])
Conf ruby-mime-types-data (3.2026.0701-1 kali-rolling [all])
Conf ruby-net-http-digest-auth (1.4.1-2 kali-rolling [all])
Conf ruby-mori (0.9.1-1 kali-rolling [all])
Conf ruby-oj (3.17.4-1 kali-rolling [amd64])
Conf ruby-rchardet (1.10.2-1 kali-rolling [all])
Conf ruby-spider (0.8.0-2 kali-rolling [all])
Conf ruby-sqlited (2.9.5-2 kali-rolling [amd64])
Conf unzip (6.0-38 kali-rolling [amd64])
Conf ruby-zip (3.4.1-dfsg-1 kali-rolling [all])
Conf samba-common (2:4.24.5+dfsg-1 kali-rolling [all])
Conf usb-tools (2:1.4.15+smbak-24.5dfsg-1 kali-rolling [amd64])
Conf tex-gyre (4.010021-8 kali-rolling [all])
Conf texlive-plain-generic (2026.20260711-2 kali-rolling [all])
Conf texlive-pictures (2026.20260711-2 kali-rolling [all])
Conf texlive-latex-base (2026.20260711-2 kali-rolling [all])
Conf texlive-latex-extra (2026.20260711-2 kali-rolling [all])
Conf texlive-latex-recommended (2026.20260711-2 kali-rolling [all])
Conf texlive-base (2026.20260711-2 kali-rolling [all])
Conf texlive-fonts-recommended (2026.20260711-2 kali-rolling [all])
Conf tpm-udev (4.1.3-7 kali-rolling [all])
Conf tumble (4.20.2-1 kali-rolling [amd64])
Conf tumble-common (4.20.2-1 kali-rolling [all])
Conf usb-modeswitch (2.6.2-2 kali-rolling [amd64])
Conf usb-ids (2026.06.26-1 kali-rolling [all])
Conf virtualbox-guest-x11 (7.2.14-dfsg-1 kali-rolling [amd64])
Conf virtualbox-guest-utils (7.2.14-dfsg-1 kali-rolling [amd64])
Conf weasyprint (69.0-7 kali-rolling [all])
Conf xbrapi (6.9.1+repack-1 kali-rolling [amd64])
Conf xdg-desktop-portal (1.22.1+deb-1 kali-rolling [amd64])
Conf xfce4-panel (4.20.8-1 kali-rolling [amd64])
Conf xfce4-settings (4.20.5-1 kali-rolling [amd64])
Conf xserver-xephyr (2:21.1.23-1 kali-rolling [amd64])
Conf xvtb (2:21.1.23-1 kali-rolling [amd64])

(kali@kali) ~/ApexPlanet/Task-4
```

7. Social Engineering — Awareness Simulation

A local training page was created as an awareness-only phishing simulation. It is explicitly labelled as a training demo and does not collect, store, or transmit credentials. The page presents common phishing warning signs and recommended user actions.



Phishing Awareness Simulation

TRAINING DEMO — NOT A REAL LOGIN PAGE

This page is part of a cybersecurity awareness exercise. Do not enter real passwords or personal information.

Can you spot the warning signs?

- Unexpected login requests
- Urgent or threatening language
- Suspicious links or domains
- Requests for passwords or sensitive information

Recommended action: Verify the sender and website independently before taking action.

Simulation complete — no credentials are collected.

8. Malware Basics

A harmless text sample was used to demonstrate basic static analysis. The file was identified as ASCII text and a SHA-256 hash was recorded. Static analysis examines artifacts without execution; dynamic analysis observes runtime behavior in an isolated environment. No malware was executed in this exercise.

Recorded SHA-256: fbe0bc72ac2e53e449c6ef65f1f6c2acc9b3fe16e1758f33d25a8535fec7f6d6

Benign sample file-type check

```
kali@kali: ~/ApexPlanet/Task-4/malware_lab

Session Actions Edit View Help

arcfour128
arcfour256
arcfour
aes192_tbc
aes256_tbc
rijndael-cbc@lysator.liu.se
aes128_ctr
aes192_ctr
aes256_ctr
mac_algorithms: (7) aes-aesni, hmac-sha1, hmac-sha256, hmac-sha512, hmac-ripemd160, hmac-ripemd256, hmac-ripemd320
hmac-md5
hmac-sha1
hmac-sha256@openssh.com
hmac-ripemd160
hmac-ripemd256@openssh.com
hmac-sha1-96
hmac-md5-96
compression_algorithms: (2)
none
zlib@openssh.com
MAC Address: 08:00:27:82:72:8E (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.77 seconds
msf exploit(mimic)(press Ctrl+C to quit) > exit

(kali@kali)-[~]
$ cd ~/ApexPlanet/Task-4
$ nano phishing_simulation.html
$ ls -l phishing_simulation.html
-rw-r--r-- 1 kali kali 814 Sep 11 06:51 phishing_simulation.html
$ xdg-open phishing_simulation.html
$ mkdir -p ~/ApexPlanet/Task-4/malware_lab
$ cd ~/ApexPlanet/Task-4/malware_lab
$ echo "ApexPlanet Task 4 - benign analysis sample" > benign_sample.txt
$ file benign_sample.txt
benign_sample.txt: ASCII text
$
```

Benign sample content and SHA-256 hash

```
kali@kali: ~/ApexPlanet/Task-4/malware_lab
Session Actions Edit View Help
aes256-ctr
mac algorithms: (7)
hmac-md5
hmac-sha1
unac-64jopenssh.com
hmac-ripemd160
hmac-ripemd160openssh.com
hmac-sha1-96
hmac-md5-96
compression_algorithms: (2)
none
zlibopenssh.com
MAC Address: 08:00:27:82:32:8E (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.17 seconds
msf exploit(multi/http/ssl_strip_session) > exit

kali@kali: ~
$ cd ~/ApexPlanet/Task-4
kali@kali: ~/ApexPlanet/Task-4
$ nano phishing_simulation.html
kali@kali: ~/ApexPlanet/Task-4
$ ls -l phishing_simulation.html
-rw-rw-r-- 1 kali kali 814 Sep 11 06:51 phishing_simulation.html
kali@kali: ~/ApexPlanet/Task-4
$ vbg-open phishing_simulation.html
kali@kali: ~/ApexPlanet/Task-4
$ mkdir -p ~/ApexPlanet/Task-4/malware_lab
kali@kali: ~/ApexPlanet/Task-4
$ cd ~/ApexPlanet/Task-4/malware_lab
kali@kali: ~/ApexPlanet/Task-4/malware_lab
$ echo 'ApexPlanet Task 4 - benign analysis sample' > benign_sample.txt
kali@kali: ~/ApexPlanet/Task-4/malware_lab
$ file benign_sample.txt
benign_sample.txt: ASCII text
kali@kali: ~/ApexPlanet/Task-4/malware_lab
$ cat benign_sample.txt
ApexPlanet Task 4 - benign analysis sample
kali@kali: ~/ApexPlanet/Task-4/malware_lab
$ sha256sum benign_sample.txt
7be0bc72ac2e53e445c6ef65f1f6c2acc903fe16e1758f33d25a8535fec7f6d6 benign_sample.txt
kali@kali: ~/ApexPlanet/Task-4/malware_lab
```

9. Risk and Mitigation Summary

Area	Evidence-based observation	Mitigation
FTP	vsftpd 2.3.4 exposed; VSFTPD backdoor module identified	Upgrade/remove; restrict exposure; secure file transfer
SSH	Legacy algorithms enumerated	Disable deprecated algorithms; update OpenSSH
Services	Multiple network services exposed	Disable unnecessary services; restrict access
Firewall	Firewall state assessed	Apply least-privilege inbound rules
Patching	Patch state assessed	Apply security updates and re-scan
Phishing	Awareness simulation created	Train users to verify senders, links and requests
Malware	Benign sample analyzed statically	Use isolated sandboxes for real samples

10. Evidence Index

The report includes the screenshots supplied for the assessment. Filenames are retained to make the report traceable to the original evidence archive.

01-Metasploitable2-Recon-Scan.png
02-Metasploit-Console.png
03-Vulnerability-Assessment.png
03-Vulnerability-Scan-Overview.png
04-Vulnerability-Finding-1.png
05-Vulnerability-Finding-2.png
06-Vulnerability-Finding-3.png
07-Vulnerability-Finding-4.png
08-SSH-Algorithm-Output.png
08-SSH-Security-Assessment.png
09-Firewall-Current-Status.png
10-Listening-Services.png
11-Patch-Assessment.png
13-Phishing-Awareness-Simulation.png
14-Benign-Sample-Static-Analysis.png
14-Metasploit-Vsftpd-Module.png
15-Benign-Sample-Analysis-Hash.png
15-Vsftpd-Module-Info.png
16-Vsftpd-Exploit-Options.png

11. Conclusion

The controlled lab assessment demonstrates a practical workflow from reconnaissance through vulnerability identification, security configuration review, awareness simulation, and basic malware-analysis concepts. The evidence most strongly supports the identification of the vulnerable VSFTPD 2.3.4 service and the need for system hardening through patching, service reduction, firewall restrictions, and modern SSH configuration.

This report intentionally distinguishes observed evidence from actions that were not successfully demonstrated in the captured screenshots.